Voronoi Homogeneity Analysis of Categorical Data

Jan de Leeuw

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TBD

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Note: This is a working manuscript which will be expanded/updated frequently. All suggestions for improvement are welcome. All Rmd, tex, html, pdf, R, and C files are in the public domain. Attribution will be appreciated, but is not required. The files can be found at https://github.com/deleeuw/voronoi

1 Introduction

1.1 Homogeneity Analysis

Homogeneity Analysis (HGA), a.k.a. *Multiple Correspondence Analysis (MCA)*, can be introduced and implemented in many different ways[1]. We refer to Guttman (1941), Tenenhaus and Young (1985), Bekker and De Leeuw (1988), Gifi (1990), Greenacre and Blasius (2006), Le Roux and Rouanet (2010), Husson et al. (2016), and De Leeuw (2023) for a bouquet of discussions and interpretations of the equations and algorithms defining the HGA technique.

A *variable* is a function that maps on a (finite or infinite) domain of *objects* into a (finite or infinite) codomain of *values*. In HGA the data are the values of m variables on the same *carrier*, a finite subset the domain consisting of n objects. Each variable j has a finite number k_j of *categories*. The categories constitute the *range* of the variable for the n objects in the carrier. Note that all variables have the same carrier. Each object maps into a unique category, and each category contains a subset of objects from the carrier. We also say that a category has a number of objects that "belong to" the category.

Here is a small example with $n = 10$ and $m = 3$, taken from Gifi (1990), page 65.

```
      [,1] [,2] [,3]
 [1,] "a"  "p"  "u"
 [2,] "b"  "q"  "v"
 [3,] "a"  "r"  "v"
 [4,] "a"  "p"  "u"
 [5,] "b"  "p"  "v"
 [6,] "c"  "p"  "v"
 [7,] "a"  "p"  "u"
 [8,] "a"  "p"  "v"
 [9,] "c"  "p"  "v"
[10,] "a"  "p"  "v"
```

The key mathematical object in HGA is the *indicator matrix*, a classical way to code categorical data[2]. For a variable j with k_j categories it is an $n \times k_j$ binary matrix G_j in which each row i contains a single element equal to one, indicating into which category the object i maps for variable j. The cross product $D_j := G_j'G_j$ is diagonal, with the *marginal frequencies* of the categories on the diagonal. For two different variables the crossproduct $C_{j\ell} = G_j'G_\ell$ is the $k_j \times k_\ell$ *cross table* or *contingency table* of the two variables. Here are the three indicator matrices of our small example, next to each other.

[1]Since PCA and MCA have three-letter acronyms I added the G of Gifi to the HA of Homogeneity Analysis.
[2]Since the indicator matrix is the core of the Gifi system we avoid the irreverent alternative "dummy coding".

```
         a b c    p q r    u v
  [1,] || 1 0 0 || 1 0 0 || 1 0 ||
  [2,] || 0 1 0 || 0 1 0 || 0 1 ||
  [3,] || 1 0 0 || 0 0 1 || 0 1 ||
  [4,] || 1 0 0 || 1 0 0 || 1 0 ||
  [5,] || 0 1 0 || 1 0 0 || 0 1 ||
  [6,] || 0 0 1 || 1 0 0 || 0 1 ||
  [7,] || 1 0 0 || 1 0 0 || 1 0 ||
  [8,] || 1 0 0 || 1 0 0 || 0 1 ||
  [9,] || 0 0 1 || 1 0 0 || 0 1 ||
 [10,] || 1 0 0 || 1 0 0 || 0 1 ||
```

We now deviate from the usual Gifi and French approaches (Husson et al. (2016)) and define HGA as a form of non-metric multidimensional scaling, using the *stress* loss function (Kruskal (1964a), Kruskal (1964b)). Define[3]

$$\sigma(X, Y_1, \cdots, Y_m, \Delta_1, \cdots, \Delta_m)) := \sum_{j=1}^{m}\sum_{i=1}^{n}\sum_{\ell=1}^{k_j} w_{i\ell}^j(\delta_{i\ell}^j - d(x_i, y_\ell^j))^2, \tag{1}$$

with $d(.,.)$ Euclidean distance.

HGA minimizes this loss over all $n \times p$ configurations X satisfying $X'X = I$, over all Y_j, and over all Δ_j that satisfy $\delta_{i\ell}^j = 0$ whenever $g_{i\ell}^j = 1$. All other elements of the Δ_j are only constrained to be non-negative.

If

$$\sigma_\star(X, Y_1, \cdots, Y_m) := \min_{\Delta_1, \cdots, \Delta_m} \sigma(X, Y_1, \cdots, Y_m, \Delta_1, \cdots, \Delta_m), \tag{2}$$

then

$$\sigma_\star(X, Y_1, \cdots, Y_m) = \sum_{j=1}^{m}\sum_{i=1}^{n}\sum_{\ell=1}^{k_j} g_{i\ell}^j d^2(x_i, y_\ell^j). \tag{3}$$

Next define

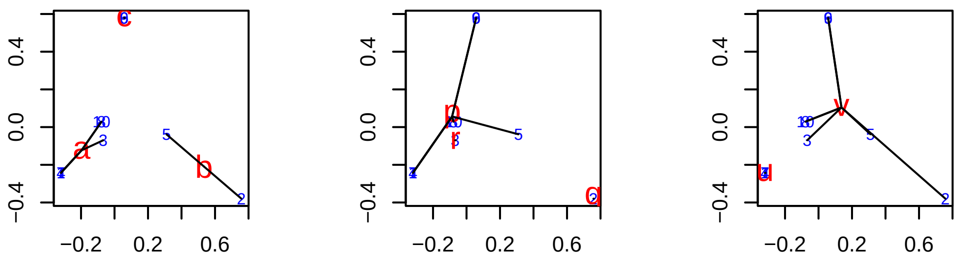
$$\sigma_{\star\star}(X) := \min_{Y_1, \cdots, Y_m} \sigma_\star(X; Y_1, \cdots, Y_m). \tag{4}$$

The minimum of (3) over the Y_j is attained at the centroids $\hat{Y}_j := D_j^{-1}G_j'X$, and if we substitute these optimal $\hat{Y}_j$ in (3) we see that $\sigma_{\star\star}(X)$ is indeed equal to the sum of squares of all lines defining the stars. Minimizing $\sigma_{\star\star}(X)$ over $X'X = I$, and thus minimizing (1) over (X, Y_j, Δ_j), defines HGA. Actually computing the solution turns out to be a straightforward

[3]The symbol $:=$ is used for definitions

eigenvalue-eigenvector problem. The details are in the publications we referenced at the beginning of this section.

A HGA in two dimensions produces of our small example the following three star plots. Objects are numbered and in BLUE, categories have bigger labels and are in RED. Note that there are ten objects, but the sets of objects $(1, 4, 7)$, $(8, 10)$, and $(6, 9)$ have the same *profile* and consequently get mapped into the same object-point. Thus there are at most six different object-points. The analysis minimizes the sum of squares of the length of all black lines in the three plots, under the condition that the matrix of object scores is column-centered and orthonormal. We can think of the results as a form of *graph drawing*, where the graph is the union of the m bipartite graphs of all m "belonging to" relations (Michailidis and De Leeuw (2001)).



1.2 Voronoi Analysis

Although using loss function (1) makes HGA a form of non-metric multidimensional scaling there are some important differences with the more familiar MDS versions minimizing stress. There is no local minima problem in HGA, because it turns out to be a symmetric eigenvalue problem. In HGA, unlike in MDS, we impose the constraint $X'X = I$. Also, and perhaps most importantly, in HGA we do not really expect the minimum loss function value to be zero, or even close to zero. The loss function measures deviations from a very unrealistic "model". Zero loss would imply all object-points coincide with all category-points they belong to. This can only be realized, except in trivial cases, if all object and category-points map into a single point, which is prohibited by the orthonormality constraint on the object scores. Thus we get a perfectly determinate and stable solution, at low computational cost, but by imposing strong constraints on the disparities and a rather arbitrary normalization. We can get around this seemingly unfortunate situation by admitting we are not fitting or testing a "model", but we are creating "best" drawings of the bipartite graphs that constitute the data.

The lack of a realistic model makes it interesting to continue to use the loss function (1), but now with the constraints that if $g^j_{i\ell} = 1$ then merely $\delta^j_{i\ell} \leq \delta^j_{i\nu}$ for all $\nu = 1, \cdots, k_j$. This is significantly weaker than the HGA constraints, which require the mn disparities $\delta^j_{i\ell}$ to be zero. If object i belongs to category ℓ of variable j we do no longer require the object-point to coincide with the category-point, but we merely require that the object-point is at least as close to category-point ℓ as to any of the other $k_j - 1$ category-points of that variable.

Voronoi Homogeneity Analysis (VHA) of categorical data minimizes loss function (1) over X, over the Y_j and over the Δ_j satisfying the relaxed ordinal constraints. It is consequently another form of non-metric (ordinal) multidimensional scaling. To be more precise, we have m linked non-metric multidimensional unfolding problems for the binary matrices G_j, which are linked because they share the same X.

Because of the different ordinal constraints on the disparities, the plots corresponding with this analysis are different. As in HGA there is a plot for each variable j with both n object-points and k_j category-points. The k_j category-points are the *generators* of k_j *Voronoi regions* that partition the space. Voronoi region $\mathcal{V}_\ell^j$ is the region of space where the points are at least as close to generator y_ℓ^j as to any of the other generators of variable j. And loss (4) is zero (for a variable) if all object-points are in the "correct" Voronoi region, i.e. in the Voronoi region of the generator corresponding with the category they belong to.

One (inefficient, but conceptually easy) way of constructing Voronoi regions in the plane is to draw the perpendicular bisectors of the (hypothetical) lines connecting all pairs of category-points of a variable. In higher dimensions $p > 2$ the bisectors are planes of dimension $p - 1$. The convex polyhedral regions thus formed, some bounded and some unbounded, define the Voronoi regions[4]. The literature on properties and computation of Voronoi regions is large, and we refer to the textbooks of Okabe et al. (2000) and Aurenhammer et al. (2013).

Since the category-point itself is obviously in its own Voronoi region and since Voronoi regions are convex it follows that if we make stars by connecting category-points with object points as in HGA then we actually require each star to be in its Voronoi region. It is no longer true that a category-point is the centroid of points of the objects belonging to the category. The category-point can even be outside the convex hull of the object-points belonging to the category.

The constraint $X'X = I$ from HGA is no longer imposed directly, but we somehow have to deal with the fact that in VHA the minimum of loss function (1) is always zero. The trivial solution, in which all points and regions collapse into a single point and all Δ_j are zero, is always available and not excluded by the constraints. But, perhaps more urgently, we have to deal with the types of trivial solutions familiar from multidimensional unfolding (Heiser (1981), De Leeuw (1983), Busing (2010)). For example, we can collapse all category-points of a variable into a single point and put all object-points on a sphere with the category-point at its center. Then all distances between category and object-points are the same and we can choose all disparities equal to that distance value and obtain loss zero.

[4] In the plane a Voronoi region is unbounded if and only if its generator is on the boundary of the convex hull of all generators.

2 Algorithm

We use an *Alternating Least Squares* or *ALS* algorithm, similar to the approach used in many other non-metric scaling methods, to minimize loss (**??**). For context, see Borg and Groenen (2005). We alternate minimization over $\Delta_1, \cdots, \Delta_m$ for given X and $Y_1 \cdots Y_m$ with minimization over X and $Y_1 \cdots Y_m$ for given $\Delta_1, \cdots, \Delta_m$. Using superscript (μ) for the iteration index we can write for an ALS iteration

$$(\Delta_1^{(\mu)}, \cdots, \Delta_m^{(\mu)}) = \underset{\Delta_1, \cdots, \Delta_m}{\text{argmin}}\ \sigma(X^{(\mu)}, Y_1^{(\mu)}, \cdots, Y_m^{(\mu)}, \Delta_1, \cdots, \Delta_m), \tag{5a}$$

$$(X^{(\mu+1)}, Y_1^{(\mu+1)}, \cdots, Y_m^{(\mu+1)}) = \underset{X, Y_1, \cdots, Y_m}{\text{argmin}}\ \sigma(X, Y_1, \cdots, Y_m, \Delta_1^{(\mu)}, \cdots, \Delta_m^{(\mu)}). \tag{5b}$$

The first substep (5a) of iteration (μ) solves a monotone regression problem for a simple partial order, in which one of k_j elements must be smaller than the other $k_j - 1$ elements. The second substep (5b) solves a metric multidimensional scaling problem. This second subproblem cannot be solved completely, because it would need an infinite iterative process by itself. Thus the second substep (5b) consists of a finite number of *inner iterations* within the second substep of a single ALS *outer iteration*.

2.1 Monotone Regression

The nm monotone regressions are applied for each object i and each variable j. Forgetting about all these indices for a moment we need to minimize the sum of squares $\|\delta - d\|^2$ over δ, where one of the elements, say δ_ν, must be smaller than or equal to all others.

This can be solved by using monotone regression with the primary approach to ties (Kruskal (1964b), De Leeuw (1977b)), treating all δ_ℓ with $\ell \neq \nu$ as ties.

2.2 Majorization

For the inner iterations we use standard *smacof* majorization steps (De Leeuw (1977a)). Because of the structure of the problem it is advantageous to decompose a majorization step into two substeps: updating X for Y_j fixed at its current values and updating the Y_j for X fixed at its current value. This could be called *Alternating Majorization*.

Let's first update X. The majorization equality we use, for current fixed Y_j and a current value $\tilde{X}$ for X, is a version of Cauchy-Schwartz.

$$d_{i\ell}^j(X, Y_j) \geq \frac{1}{d_{i\ell}^j(\tilde{X}, Y_j)}(x_i - y_\ell^j)'(\tilde{x}_i - y_\ell^j). \tag{6}$$

If we define

$$b_{i\ell}^j(\tilde{X}, Y_j) := w_{i\ell}^j \frac{\delta_{i\ell}^j}{d_{i\ell}^j(\tilde{X}, Y_j)} \tag{7}$$

then

$$\begin{aligned}\sigma_j(X, Y_j) \leq C_j - 2\sum_{i=1}^n \sum_{\ell=1}^{k_j} b_{i\ell}^j(\tilde{X}, Y_j)(x_i - y_\ell^j)'(\tilde{x}_i - y_\ell^j) \\ + \sum_{i=1}^n \sum_{\ell=1}^{k_j} w_{i\ell}^j (x_i - y_\ell^j)'(x_i - y_\ell^j),\end{aligned} \tag{8}$$

where C_j does not depend of X. The right-hand-side of (8) must be minimized over X. Use the symbol $\star$ to replace a subscript or superscript over which we have summed. The stationary equations are

$$w_{i\star}^\star x_i = b_{i\star}^\star(\tilde{X}, Y_j)\tilde{x}_i - \sum_{j=1}^m \sum_{\ell=1}^{k_j} (b_{i\ell}^j(\tilde{X}, Y_j) - w_{i\ell}^j) y_\ell^j. \tag{9a}$$

In the same way we can update $\tilde{Y}_j$. By symmetry

$$w_{\star\ell}^j y_\ell^j = b_{\star\ell}^j(X, \tilde{Y}_j)\tilde{y}_\ell^j - \sum_{i=1}^n (b_{i\ell}^j(X, \tilde{Y}_j) - w_{i\ell}^j) x_i \tag{9b}$$

3 Constraints

3.1 Regression Constraints

In this paper we use the constraint

$$\sum_{\ell=1}^{k_j}(\delta_{i\ell}^j-\overline{\delta}_{ij})^2=c_j, \tag{10}$$

with $\overline{\delta}_{ij}$ the mean of the $\delta_{i\ell}^j$ and with c_j constants. For instance we could choose $c_j=1$ or $c_j=k_j$ for all j. In addition, of course, the $\delta_{i\ell}^j$ must satisfy the ordinal constraints. Constraint (10) explicitly excludes the possibility that the $\delta_{i\ell}^j$ for given i and j are equal for all ℓ.

Consider the cone of all vectors δ in $\mathbb{R}^n$ for which $\delta_i\leq\delta_\ell$ for all $i\neq\ell$ and $e'\delta=0$. The extreme rays of the cone will be the $n-1$ vectors of the form $\alpha e_i+\beta(e-e_i)$ with $i\neq\ell$. Thus element i is equal to α and all other elements (including δ_ℓ) are equal to β. Thus we must have $\alpha>\beta$. The sum of the elements is $\alpha+(n-1)\beta$, which must be zero. Thus $\alpha=-(n-1)\beta$, which implies $\beta<0$ and $\alpha>0$. The extreme rays are positive multiples of the vectors $e_i-n^{-1}e$ with $i\neq\ell$ and the cone is the set of all weighted sums with non-negative weights of these $n-1$ vectors. Here is a small example.

```r
a <- -cbind(1, -diag(4))
b <- (diag(5)-(1 / 5))[, -1]
print(a)
```

```
     [,1] [,2] [,3] [,4] [,5]
[1,]   -1    1    0    0    0
[2,]   -1    0    1    0    0
[3,]   -1    0    0    1    0
[4,]   -1    0    0    0    1
```

```r
print(b)
```

```
     [,1] [,2] [,3] [,4]
[1,] -0.2 -0.2 -0.2 -0.2
[2,]  0.8 -0.2 -0.2 -0.2
[3,] -0.2  0.8 -0.2 -0.2
[4,] -0.2 -0.2  0.8 -0.2
[5,] -0.2 -0.2 -0.2  0.8
```

```r
print(a %*% b)
```

```
     [,1] [,2] [,3] [,4]
[1,]    1    0    0    0
[2,]    0    1    0    0
[3,]    0    0    1    0
[4,]    0    0    0    1
```

For later reference the sum of squares of the vectors $e_i - n^{-1}e$ is $(n-1)/n$, so the vectors $\sqrt{n/(n-1)}(e_i - n^{-1}e)$ are the vectors of length 1 on the extreme rays.

Suppose y is a vector with n elements. The problem we study is to minimixe the sum of squares $\sigma(x) := \|x-y\|^2$ over $x \in K$ and $x'Jx = 1$. Here

- K is the cone of vectors with $x_1 \leq x_2 \leq \cdots \leq x_n$,
- J is the centering matrix $J := I - n^{-1}ee'$, where
- e is the vector with all n elements equal to one.

Change variables to $\tilde{x} = Jx$. Then $x'Jx = \tilde{x}'\tilde{x}$ and $x \in K$ if and only if $\tilde{x} \in K$. Also $\tilde{x} = Jx$ if and only if $x = \tilde{x} + \beta e$. So the transformed problem is to minimize $\|\tilde{x} + \beta e - y\|^2$ over β and $\tilde{x} \in K$, with $e'\tilde{x} = 0$ and $\tilde{x}'\tilde{x} = 1$. The minimizer for β is $\overline{y} := n^{-1}e'y$, and thus the problem becomes minimizing $\|\tilde{x} - Jy\|^2$ over $\tilde{x} \in K$, $\tilde{x}'\tilde{x} = 1$, and $e'\tilde{x} = 0$.

Forget about the constraint $e'\tilde{x} = 0$. If M is the monotone regression operator it follows that the solution for $\tilde{x}$ is $M(Jy)/\|M(Jy)\|$, which automatically satisfies $e'\tilde{x} = 0$. Thus the solution of our original problem is

$$\hat{x} = \frac{M(y-\overline{y})}{\|M(y-\overline{y})\|} + \overline{y}.$$

$M(y-\overline{y}) = M(y) - \overline{y}$ and thus $\|M(y-\overline{y})\|^2 = \|M(y)\|^2 + n\overline{y}^2 - 2\overline{y}e'M(y)$ nor $e'M(y) = e'y = n\overline{y}$. Thus $\|M(y-\overline{y})\|^2 = \|M(y)\|^2 - n\overline{M(y)}^2$

$$\hat{x} = \frac{M(y) - \overline{y}e}{\|M(y-\overline{y})\|} + \overline{y}e = \frac{1}{\|M(y-\overline{y})\|}M(y) + \left\{1 - \frac{1}{\|M(y-\overline{y})\|}\right\}\overline{y}e$$

3.2 Configuration Constraints

If there are constraints on X and/or Y we use majorization theory for restricted MDS(from (De Leeuw and Heiser (1980)). Suppose

$$\tilde{Z} := \begin{bmatrix}\tilde{X}\\ \tilde{Y}\end{bmatrix}$$

is the unrestricted Guttman transform. In the majorization step we minimize (or decrease)

$$\text{tr } (Z - \tilde{Z})'V(Z - \tilde{Z}) = m\|X - \tilde{X}\|^2 + n\|Y - \tilde{Y}\|^2 - 2\text{ tr } (X - \tilde{X})'ee'(Y - \tilde{Y}) \quad (11)$$

over X and Y satisfying the constraints.

3.2.1 Column-Centering

If we require $X'e = 0$ then the update becomes $J_n\tilde{X}$, with $J_n = I - n^{-1}ee'$ the centering matrix.

This is different from updating X as $J_n\tilde{X}$ and Y as $\tilde{Y} - n^{-1}ee'\tilde{X}$. With this alternative update, in which we subtract the column-means of $\tilde{X}$ from both $\tilde{X}$ and $\tilde{Y}$, the distances are the same before and after centering. Thus the distances in all iterations, and consequently the loss function values, are the same with or without this type of centering.

In general it is advantagous to require $e'X = 0$ because the third term in equation (11) becomes independent of X, and thus the (X, Y) minimization step is separable (can be solved for X and Y separately).

3.2.2 Orthogonality

If we require $X'X = I$, together with $e'X = 0$, as in HGA, the solution for X is KL', where $J_n\tilde{X} = K\Lambda L'$ is a singular value decomposition.

3.2.3 Rank Constraints

One of the main themes in Gifi (1990) is that the $k_j \times p$ matrices of category quantifications can be constrained to be of rank $r < p$. We will restrict ourselves in this paper to $r = 1$, which defines what Gifi calls *single quantifications*. If all variables are restricted to have single quantifications then a HGA becomes equivalent to a PCA of the transformed variables.

In VHA the Voronoi regions of single variables become parallel strips. The category-points of a variable are on a line through the origin, and the Voronoi lines or planes are parallel and perpendicular to the line with the category-points.

$$-2\text{tr }(X-\tilde{X})'ee'(Y-\tilde{Y})+n\|Y-\tilde{Y}\|^2=\text{tr }\|(Y-\tilde{Y})-n^{-1}ee'(X-\tilde{X})\|^2+C$$

So rank one approximation of $\tilde{Y}+n^{-1}ee'(X-\tilde{X})$. If X is column-centered then of course $\tilde{Y}-ee'\tilde{X})$.

3.2.4 Centroid

A more complicated restriction is $Y=D^{-1}G'X$, possibly together with $X'X=I$ and $e'X=0$. The *centroid constraints* effectively eliminate the Y_j from the optimization problem. We have to adapt the majorization method to take the constraints into account.

Consider a single variable j and the corresponding $n\times k_j$ distance matrix $D_j(X)$. We have $d_{i\ell}(X)=\|t\|$, where $t:=X'r$ and $r:=e_i-d_l^{-1}g_l$. Now

$$d_{i\ell}=\sqrt{t't}\geq\frac{1}{\sqrt{\tilde{t}'\tilde{t}}}\tilde{t}'t=\frac{1}{\sqrt{\tilde{t}'\tilde{t}}}\tilde{t}'X'r=\text{tr }X'\frac{r\tilde{t}'}{\sqrt{\tilde{t}'\tilde{t}}},\tag{12a}$$

$$d_{il}^2=t't=\text{tr }X'rr'X,\tag{12b}$$

with $\tilde{t}:=\tilde{X}'r$.

The loss component for the variable is

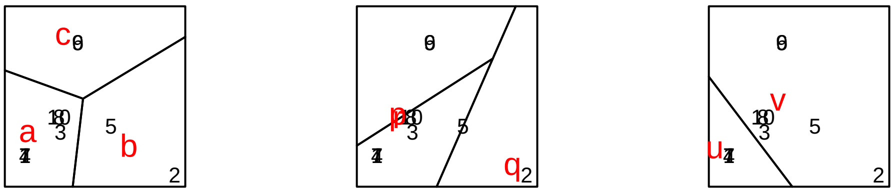
$$\sigma_j(X)=\sum_{i=1}^n\sum_{\ell=1}^{k_j}w_{i\ell}(\delta_{i\ell}-d_{ij}(X))^2$$

4 Software

5 Examples

5.1 Small Example

We start with the small example we used before to illustrate HGA.



Loss is 0.0000009581 after 33 iterations. If we compare the plots from the Voronoi analysis with the star plots from the initial HGA for this example we see that the algorithm did not have to change much to obtain perfect fit. One thing that does change is that the iterations moves the category-points “p” and “r” for variable 2 very close to each other. At the point where we stop the iterations they are still distinguishable, and thus their perpendicular bisector is still a line in the Voronoi plot. If we would continue iterating “p” and “r” would coincide and there would only be two Voronoi regions: one containing category-point “q” and object-point 2, and one containing all other points. one containing all other

5.2 GALO

Loss is 0.0008972348 after 1000 iterations.

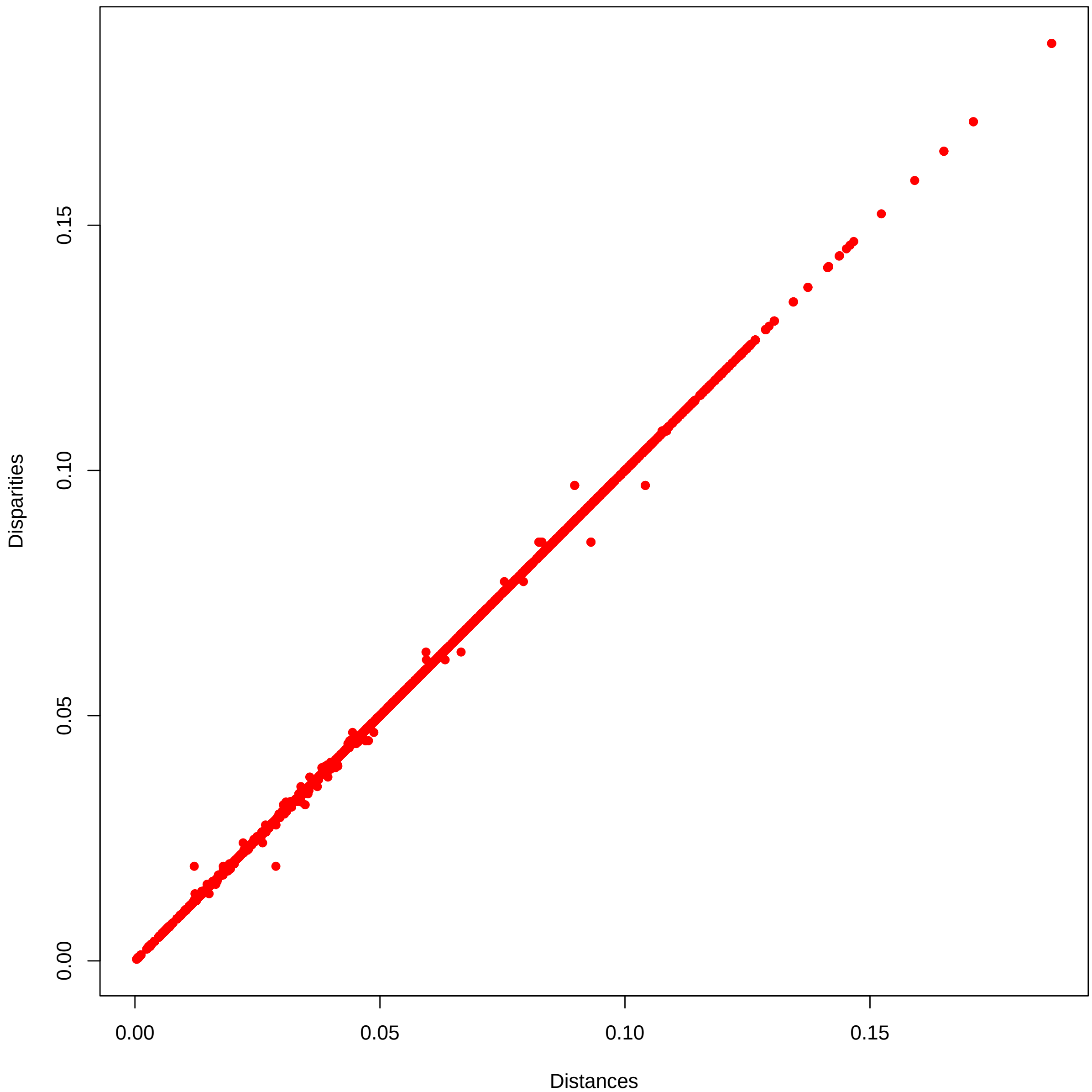


Figure 1: Galo D-Dhat Plot

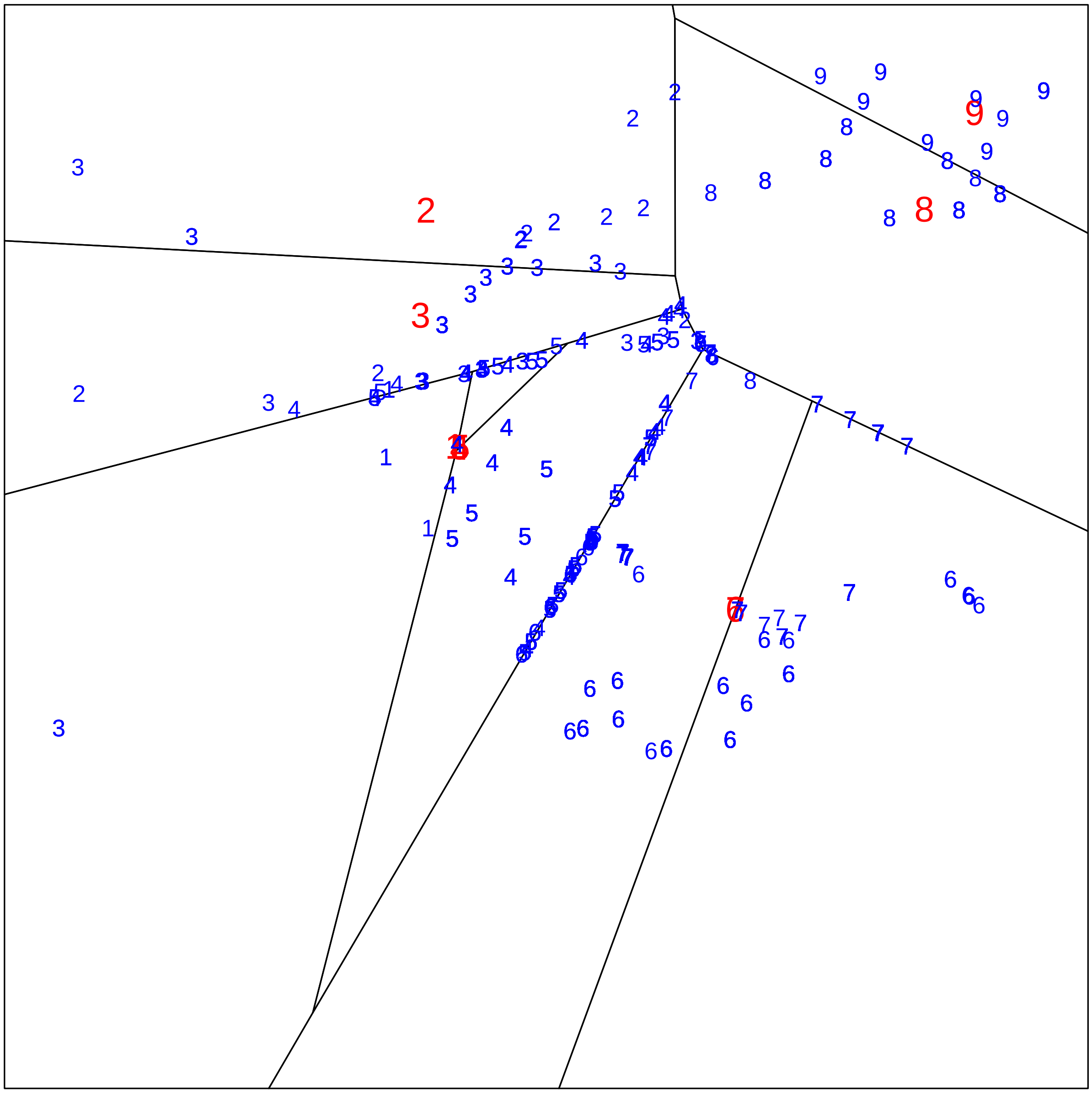


Figure 2: Galo IQ

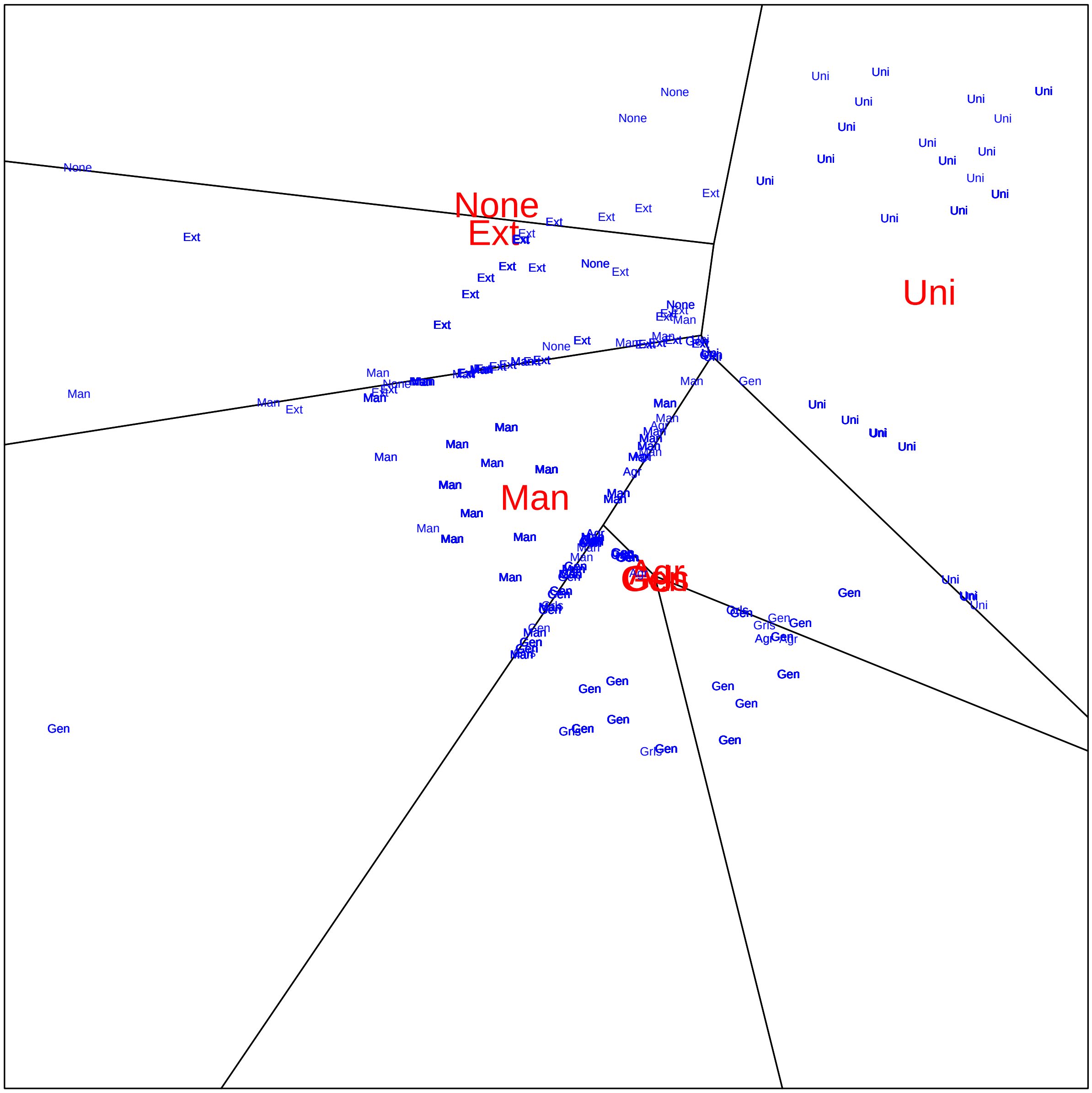


Figure 3: Galo ADV

6 Appendix: Majorized Alternating Least Squares (MALS)

Consider the following problem: minimize the loss function

$$\sigma(X,Y) := \sum_{i=1}^{n}\sum_{j=1}^{m} w_{ij}(\delta_{ij} - d(x_i,y_j))^2$$

over X in a subset $\mathcal{X}$ of $\mathbb{R}^{n\times p}$ and Y in a subset $\mathcal{Y}$ of $\mathbb{R}^{m\times p}$. Here $d(x_i,y_j)$ is the Euclidean distance between row i of X and row j of Y. The weights w_{ij} and the dissimilarities δ_{ij} are given non-negative numbers.

The algorithm we propose in this appendix alternates between decreasing loss by modifying X for fixed Y and decreasing loss by modifying Y for fixed X. This is the alternating least squares (ALS) component of our algorithm. Majorization is applied to both the function of X minimized in the first ALS step and to the function of Y minimized in the second ALS step.

We derive the update formula for X, the one for Y follows by symmetry. First expand stress

$$\sigma(X,Y) = C - 2\ \sum_{i=1}^{n}\sum_{j=1}^{m} w_{ij}\delta_{ij}d(x_i,y_j) + \sum_{i=1}^{n}\sum_{j=1}^{m} w_{ij}d_{ij}^2(x_i,y_j)$$

Here C is used as a generic term that does not depend on X. So C can mean different things in different formulas.

Suppose $\tilde{X}$ and $\tilde{Y}$ are our current values of X and Y. Then by the CS-inequality

$$d(x_i,\tilde{y}_j) \geq \frac{1}{d(\tilde{x}_i,\tilde{y}_j)}(x_i-\tilde{y}_j)(\tilde{x}_i-\tilde{y}_j)$$

Define

$$\tilde{b}_{ij} := w_{ij}\frac{\delta_{ij}}{d(x_i,y_j)}$$

Then

$$\sigma(X,\tilde{Y}) \leq \eta(X,\tilde{X},\tilde{Y})$$

with

$$\eta(X,\tilde{X},\tilde{Y}) := C - 2\ \sum_{i=1}^{n}\sum_{j=1}^{m}\tilde{b}_{ij}(x_i-\tilde{y}_j)'(\tilde{x}_i-\tilde{y}_j) + \sum_{i=1}^{n}\sum_{j=1}^{m} w_{ij}(x_i-\tilde{y}_j)'(x_i-\tilde{y}_j)$$

Now

$$\sum_{i=1}^{n} x_i' \sum_{j=1}^{m}\tilde{b}_{ij}(\tilde{x}_i-\tilde{y}_j) = \text{tr}\ X'(\tilde{B}_r\tilde{X} - \tilde{B}\tilde{Y}) + C$$

with $\tilde{B}_r$ the diagonal matrix of row sums of $\tilde{B}$. Also

$$\sum_{i=1}^n\sum_{j=1}^m w_{ij}(x_i-\tilde{y}_j)'(x_i-\tilde{y}_j)=\text{tr }X'W_rX-2\text{ tr }X'W\tilde{Y}+C$$

with W_r the diagonal matrix of row sums of W. Gathering terms gives

$$\eta(X,\tilde{X},\tilde{Y})=\text{tr }(X-\overline{X})'W_r(X-\overline{X})+C$$

with

$$\overline{X}=W_r^{-1}(\tilde{B}_r\tilde{X}-(\tilde{B}-W)\tilde{Y})$$

By symmetry

$$\eta(Y,\tilde{X},\tilde{Y})=\text{tr }(Y-\overline{Y})'W_c(Y-\overline{Y})+C$$

with

$$\overline{Y}=W_c^{-1}(\tilde{B}_c\tilde{Y}-(\tilde{B}-W)'\tilde{X})$$

using the diagonal matrices W_c and B_c of column sums.

The algorithm to minimize stress is thus as follows. Our current best values in iteration k are $X^{(k)}$ and $Y^{(k)}$.

$$X^{(k+1)}=\text{proj}_r(W_r^{-1}(\tilde{B}_r^{(k,k)}X^{(k)}-(B^{(k,k)}-W)Y^{(k)}))$$

$$Y^{(k+1)}=\text{proj}_c(W_c^{-1}(\tilde{B}_c^{(k+1,k)}Y^{(k)}-(B^{(k+1,k)}-W)'X^{(k+1)}))$$

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